

Ethan Villanueva

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SUMMARY

Computer Science undergraduate at the University of Nottingham, projected First-class honours, following a 13-month industry placement at **Siemens Digital Industries Software**. Experienced in the end-to-end development lifecycle within a large-scale C#/.NET enterprise environment, including feature delivery, automated testing, and CI/CD pipeline management via Azure DevOps.

TECHNICAL SKILLS

- **Languages:** C#, Python, Java, C, SQL, JavaScript, HTML, CSS
- **Frameworks:** .NET, WPF/MVVM, NuGet, Scikit-Learn, Pandas, Matplotlib, JUnit
- **Databases:** SQL Server, PostgreSQL, AWS DynamoDB, SQLite, Supabase
- **Cloud/DevOps:** Azure DevOps, CI/CD Pipelines, YAML, PowerShell, AWS, Git
- **Testing:** Unit, Integration, Regression, Automated UI Testing, Playwright, JUnit
- **Systems:** Multi-threading, POSIX Threads, TCP/IP, Linux/Unix

EXPERIENCE

Siemens Digital Industries Software

Jul 2025 - Aug 2026

Software Engineer Intern

Milton Keynes

- Built and shipped product features across the full stack of the Simcenter FLOMASTER codebase, tracing changes through **WPF/MVVM** UI, C# simulation logic, **SQL** Server stored procedures, and **NuGet** package dependencies across 7 repositories.
- Automated SDK documentation delivery end-to-end: designed **Azure DevOps** YAML pipelines and **PowerShell** scripts to migrate legacy CHM help files to browser-based HTML, integrating output into continuous-build, installer, and nightly consumers.
- Diagnosed a critical UI-test VM regression by comparing environments and tracing a severe test slowdown to a **SQL Server 2025** incompatibility; restored expected test performance by reconfiguring the affected VM with **SQL Server 2022**.
- Maintained and expanded automated test coverage across unit, integration, regression, and **automated UI** test suites, writing tests as part of every feature or fix to prevent regressions reaching release.
- Onboarded and trained incoming interns on **C#**, **MVVM**, and **NuGet**; conducted pull request reviews and delivered structured handover documentation.

Capital One

Oct 2024 - May 2025

Back-end Developer

Remote

- Designed **AWS DynamoDB** schemas and mock **REST APIs** to simulate banking transaction and account notification services for a Capital One-sponsored Customer Communication Channel, working in an 8-person agile team.
- Researched and applied **GDPR** data privacy controls to ensure the system architecture adhered to financial data compliance requirements, documenting decisions as formal acceptance criteria.
- Collaborated with Capital One engineers across regular agile sprints, defining user stories, acceptance criteria, and API contracts to align the university-side implementation with industry requirements.

PROJECTS

Mindmaply - LeedsHack 2026 Winner: Best Use of Google Gemini API (Python, Gemini API, Vector Embeddings)

- Engineered the core **Python** conversation engine and **graph data models** to support non-linear LLM conversation branching across dynamically created topic nodes.
- Implemented intent detection using **vector embeddings** and **Gemini API** prompts to automatically branch conversations when subjects shift, without user intervention.
- Built context scoping logic and automated background summarisation to pass only minimal, relevant context to the **Gemini API** — preventing model drift across long sessions.
- Handled state orchestration and integrated backend graph traversal with the frontend UI.

Operating System Simulator (C, POSIX Threads, Linux/Unix)

- Built an optimised **round-robin scheduler** to manage multi-threaded task queues with fairness and efficiency across concurrent processes.
- Implemented **POSIX threads**, **mutexes**, and **semaphores** to handle concurrency and prevent race conditions.
- Verified reliability by eliminating deadlocks and memory leaks through rigorous testing with **Valgrind**.

Forest Fire Prediction Models (Python, Scikit-Learn, Pandas, Matplotlib)

- Built linear regression and **neural network** predictive models to predict forest fire burn areas from meteorological data.
- Evaluated models with **k-fold cross-validation**, achieving a Mean Squared Error of **0.058** with the neural network model.
- Visualised dataset insights, regression residuals, and model performance in Jupyter Notebook using **Matplotlib**.

Route Network Server (C, TCP/IP, Berkeley Sockets)

- Engineered a multi-client **TCP/IP** server to query and manage network graphs over persistent socket connections.
- Implemented **Dijkstra's Algorithm** over a custom graph library to calculate optimal transit paths in real time.
- Designed a custom client-server protocol supporting dynamic node and edge updates with persistent connection state.

Vehicle Management System (HTML, CSS, JavaScript, PostgreSQL, Supabase)

- Built a responsive full-stack web application managing vehicle and owner records via a **Supabase PostgreSQL** backend with structured relational tables.
- Implemented **RESTful API** interactions for user queries and automated end-to-end integration tests using **Playwright**.
- Audited performance and accessibility with **Google Lighthouse**.

Test-Driven Development Group Project (Java, JUnit)

- Led a team of 3 in refactoring a legacy property viewing application, applying **red-green-refactor TDD** cycles and writing comprehensive automated tests in **JUnit** throughout.
- Achieved **96.8%** and **94.0%** code coverage across the two primary domain classes.
- Decoupled tightly linked legacy components into clean, modular classes following **SOLID** design principles.

EDUCATION

University of Nottingham

Sep 2023 - Jun 2027

Bsc Computer Science with Year in Industry

- Projected First-class honours
- Relevant modules: Developing Maintainable Software, Databases and Interfaces, Operating Systems & Concurrency, Systems and Architecture, Artificial Intelligence Methods, Mathematics for Computer Scientists, Programming and Algorithms, Programming Paradigms, Software Engineering Group Project

King Edward VI College

Sep 2020 - Jun 2022

A-Levels

- Achieved AAA in Physics, Maths, Geology.

Queen Mary's Grammar School

Sep 2015 - Jun 2020

GCSEs

- Obtained All 7's/8's (A/A*) across 11 subjects including IT, Maths, Physics, and English.

CERTIFICATIONS/AWARDS

- **AWS Certified Cloud Practitioner** - Amazon Web Services
- **LeedsHack 2026 Winner: Best Use of Google Gemini API** - Major League Hacking